The bandwidth of the floor-attaining census, corrected on a lattice; the price of a dial-blind birth

Supporting computations for arXiv:2608.18886 — deposit note, version 4

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Correction to version 3

The band-limited prices of version 3 were computed on 1024 random offsets. A census band-limited to K harmonics evaluated on N random atoms computes its correlators as an N-point random quadrature (error $\sim N^{-1/2}$), and the LP spends coefficients cancelling that noise to hold the 1% band, at a cost in MD. On 4096 random offsets every low-K price is lower; on a 32×32 lattice (exact quadrature below the 16th harmonic) the effect is absent. Version 3's values are correct for its 1024-random census and are superseded as the model's smoothness price by the lattice values below.

Results ($\varepsilon = 0.01$; MD/floor)

Floor attained on every sampling (1.000000: 1024 random $\times 4$, 4096 random, 1024 lattice).

K	**1024 random (v3)**	**4096 random**	**1024 lattice**
6	2.274	1.828	**1.700**
8	1.861	1.428	**1.429**
10	1.598	1.301	**1.258**
12	1.356	1.219	**1.112**
14	1.121	1.139	**1.000**
16	—	1.071	1.000
18	—	1.012	—

Minimum miss of the best K-band census (certified LP minima): $K = 3$: 10.5 / 12.6 / 8.5%; $K = 4$: 1.74 / 1.43 / 1.17%; $K = 5$: 1.20 / 1.03 / **1.055%** (1024 random / 4096 random / lattice). $K = 2$: no non-negative census satisfies the CHSH row.

Statements

(i) No census band-limited below the sixth harmonic reproduces the singlet to 1% on this terrain, on any sampling; the margin is thin (a $K = 5$ census reaches 1.05%). (ii) The smoothest attaining census pays 1.70× the floor on the lattice; the price falls to the floor by $K = 14$ on the lattice and by $K \approx 18$–20 on random offsets: the floor census carries angular structure of 5–6°. (iii) The census's skeleton is sampling-independent: on the lattice, (2,2) at $\varphi = 159°$ (random: 157°), preferred sum 39.8° mod 90° (random: 39.5°), the same (S, D) cross pixels and both families of push; the station asymmetry is on the same side.

Dial dependence and the dial-blind price

The floor-attaining census is not the same offset density at every (τ, scale) cell: the best cell-separable approximation of the smooth $K = 6$ census misses the singlet by 5–6%, and cell shapes are more alike within a scale (cosine 0.70) than within a τ (0.43). The class of dial-blind censuses — one or two offset densities shared by all cells, reweighted per cell, shapes free over the atoms — reproduces the singlet (miss 0.4% / 0.08%) but pays **1.47× / 1.20×** the floor on the lattice (1.8 / 1.6 on 1024 random, with the quadrature cost). A census required to be both dial-blind and smooth was not found under a 1.2% miss at $K = 6$ or 8 with one shape on either sampling; with two shapes it reaches the band at $K = 6$ at 2.37×; with one shape at $K = 10$ at 2.27× (alternating LP, six starts; best found, not certified).

Parametric birth model

A ~30-parameter von Mises push model with a preferred axis (cell-separable by construction) reaches a 2.4–2.6% miss cold or warm-started from the $K = 6$ census, which it fits with 0.70 relative error; it finds the axis (38°), the station slots (38° mod 90°) and the (2,2) phase (156°) but lacks the high-D harmonics the free dial-blind shape uses. A birth model that attains the floor must carry structure to $K \approx 14$–18 and vary with the terrain depth.

Scope

One terrain (the paper's); six censuses (four 1024-random, one roles-swapped, one 4096-random, one lattice); LP prices and $K \leq 5$ misses are exact optima; the dial-blind and birth-model results are local optima reported as found.

Files

Scripts `pipeline_4096_v2.py`, `census_cells_v1.py`, `census_cellsep_v1.py`, `census_cellsep_v2.py`, `birth_model_v1.py`, `birth_model_v2.py`, `run_parallel_v4.py`–`v8.py` and their launchers; transcripts `pipeline_4096_console.txt`, `pipeline_1024_console.txt`, `bl_n64_k*.txt`, `bl_n64_k*_miss.txt`, `bl_n32_k*.txt`, `census_structure_v2_nsh32_grid0.txt`, `census_sumdiff_v1_nsh32_grid0.txt`, `census_cells_*.txt`, `census_cellsep_*.txt`, `cs_*.txt`, `bm_*.txt`, `bm2_*.txt`, `bl_r0_e*_k4.txt`, `bl_r0_e*_k5.txt`, `parallel4–8_console.txt`; banks and censuses for 4096 random and the lattice; the `census_bandlimit_v4_*`, `census_cellsep_*`, `birth_model_*` and grid structure JSONs. `pipeline_4096_v1.py` is withdrawn (no `__main__` guard; produced no output) and is not deposited.